

The empirical recurrence for OEIS A234229: recovering a conjecture that is not written down

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Abstract

OEIS A234229 records “Empirical recurrence of order 40”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 512$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 40, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 40 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234229 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 3 (constant-stress 1×1 tilings)”. It has offset 1 and begins

5260, 57912, 635904, 7024396, 77499592, 859364784, 9519608332, 106004400596, ...

The entry states:

Empirical recurrence of order 40 (see link above).

As of the “Last modified” line on the live entry (Jun 19 2022 23:40:31, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 512$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1024$ exact terms of the model returns order 40 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{40} c_i a(n-i),$$

c_1	35	c_{11}	−389142955323	c_{21}	−145846761811958300832	c_{31}	4404436681418864
c_2	297	c_{12}	91724981611836	c_{22}	32785662669701112192	c_{32}	−3360606601979817
c_3	−23436	c_{13}	−197800137133940	c_{23}	1289945293647306361152	c_{33}	7800367383592862
c_4	87922	c_{14}	−3712235714726180	c_{24}	−1184349201595312850304	c_{34}	36567116095001370
c_5	6354341	c_{15}	14345603246285440	c_{25}	−7442731504916052238080	c_{35}	−2168496592542920
c_6	−57467833	c_{16}	91658915774466240	c_{26}	10968787647661156327680	c_{36}	−1849731725838016
c_7	−868697509	c_{17}	−507226585917587320	c_{27}	27228854162118940992000	c_{37}	18615991777907830
c_8	12179862085	c_{18}	−1321387528802707232	c_{28}	−56237696679658175631360	c_{38}	9747586176417792
c_9	56717124080	c_{19}	10762321399020285120	c_{29}	−57389962891214459289600	c_{39}	−5313800827463270
c_{10}	−1376288977822	c_{20}	8733841242134807264	c_{30}	176042457898795709798400	c_{40}	1421387052613632

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 40, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $40 + 4$ terms removed returns order 40 as well, so $\deg q_{\min} = 40 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $40 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 40 that this sequence satisfies — and that family turns out to have exactly one member.

References

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